

Assignment Brief: Mini Project Planner with Gantt View

Build a simple project planning tool where a user can:

- Add tasks with durations and dependencies (just precedence dependency)
 - Auto-calculate a **basic schedule**
 - View the tasks in a Gantt chart format
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Tech Stack

- MongoDB
 - Express.js + Node.js
 - React
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Core Features

1. **Create Tasks (If UI can be created, its great, else just create good data model and add the tasks from backedn with that data model)**
 - Task name
 - Duration in days
 - Dependencies
 2. **Schedule Tasks (create API which will create the schedule with these tasks)**
 - Based on the dependencies, calculate:
 - Earliest start date
 - End date = start date + duration
 - Tasks without dependencies start on **Day 0**
 3. **Display in Gantt Chart**
 - Each task appears as a horizontal bar
 - X-axis: timeline
 - Y-axis: task names
 - Show duration and start time
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Example Task Data

Task ID	Name	Duration	Depends On
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1	Design	3 days	None
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2	Frontend	4 days	1
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3	Backend	5 days	1
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4	Testing	2 days	2, 3
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→ Expected schedule:

- Task 1 (Design): Day 0–3
- Task 2 & 3: start after Day 3
- Task 4: starts after both 2 & 3 finish → after Day 8